Venkata Lakshmi Parimala Pasupuleti

Applications Developer · Enterprise Database Systems · ETL & Data Pipelines · System Integrations & APIs · Reporting & Analytics · 9yr Production

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PROFESSIONAL SUMMARY

Applications developer and data systems specialist with 9 years owning enterprise data pipelines, database systems, and multi-platform integrations in a compliance-critical environment. At Entrata, I was the long-term sole owner of complex data import workflows processing millions of regulatory records annually -- managing data mapping, transformation, validation, deduplication, and quality controls for systems used by 100K+ users. I design and maintain integrations between internal platforms and third-party vendors, build custom reporting and dashboards for business stakeholders, and define data governance standards that keep enterprise systems reliable over time. I work closely with operations, compliance, and business teams to translate requirements into technical solutions. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Portfolio: venkatalakshmiparimala.github.io/Portfolio/about.html

TECHNICAL SKILLS

Database & Data Engineering: SQL/PL-SQL/PL-SQL (9yr production), PostgreSQL (200+ query optimization, stored procedures, complex schemas, multi-tenant environments), Oracle (familiarity; PL/SQL transferable), MySQL, Microsoft SQL Server (T-SQL), ETL/ELT Pipelines, Data Modeling, Data Mapping & Transformation, Data Validation & Quality Controls, Deduplication, Data Cleansing, Data Governance, Reporting Frameworks, Automated Data Integrity Checks, Documentation & SOPs

Reporting, Analytics & Visualization: Tableau, Power BI, Python (Pandas, NumPy, Matplotlib, Seaborn, Plotly), R (ggplot2, statistical analysis), Jupyter, Dashboard Design, Ad-hoc Reporting, Stakeholder Analytics Delivery

Development, Integration & Web: Python (9yr primary), JavaScript, HTML/CSS, SQL scripting, Bash/shell scripting, REST APIs, API Integration, System Integrations, Third-Party Data Synchronization, AWS (S3, EC2, RDS, Glue), Docker, CI/CD (GitHub Actions), Git, Agile/Scrum; PHP (familiarity), XML; Security & Access Controls, Data Privacy, HIPAA-compliant systems

EXPERIENCE

Entrata -- Multifamily Real Estate SaaS (PropTech)

Sep 2018 -- Aug 2024

Senior Software Engineer -- Enterprise Data Systems, Integrations & Compliance Reporting · Lehi, UT (Remote)

- **Long-term owner of enterprise data import and management pipelines** -- sole owner of Entrata's annual HUD income-limit import: complex federal regulatory data (thousands of source records transformed and mapped to ~180,000 domain records) powering eligibility, rent calculations, and tax-credit compliance for 100K+ housing units across multi-tenant environments. Defined data mapping standards, transformation rules, and validation logic; conducted pre-import data audits; managed deduplication and reconciliation; maintained rollback capability. Zero data integrity violations across full 7-year ownership period.
- **Multi-platform integrations and third-party data synchronization** -- built and maintained integrations between internal CRM/platform systems and external vendor applications; managed data import pipelines from internal and third-party sources; established monitoring, alerting, and error-resolution protocols; defined standards for data exchange and synchronization across platforms. AWS S3 migration with encrypted ETL pipelines (100% PHI compliance, zero security incidents).
- **Reporting, dashboards, and stakeholder analytics** -- built reporting frameworks for compliance and operational data; developed and maintained complex SQL queries, stored procedures, and automated scripts; delivered ad-hoc and scheduled reports to operations, compliance, and business leadership teams; optimized 200+ SQL queries (30% performance gain) through execution plan analysis, indexing, and stored procedure refactoring.
- **Data governance, security, and access management** -- managed user access controls and role-based permissions across multi-tenant data systems; enforced data security standards protecting sensitive resident and compliance data; maintained Libsodium-encrypted pipelines for PHI; implemented data quality controls and documented standard operating procedures for institutional knowledge retention.
- **Cross-functional collaboration and mentorship** -- partnered directly with operations, compliance, and customer success teams to translate business requirements into technical solutions; mentored 5+ engineers; conducted 300+ peer code reviews; 99.5% uptime across production systems.

George Mason University

Aug 2024 -- May 2026

Research Associate & Graduate Teaching Assistant

- Built multi-source data pipelines and prediction models (Python, R) across 190+ countries (R^2=0.977); applied Isolation Forest to automate data integrity checks across 3 research pipelines; designed Tableau and ggplot2 dashboards improving research comprehension 40% for faculty leadership; delivered data analytics instruction to 600+ students.

Magizhs Tech Solution

May 2025 -- Aug 2025

AI/ML Engineer -- Data Systems & API Integration

- Designed and built data retrieval and validation pipelines with multi-step API integration; implemented data transformation and quality rules; established automated validation scripts enforcing data integrity before export.

Entrata India -- Pune

Jan 2017 -- Sep 2018

Associate Software Engineer -- Database Systems & Compliance Data Workflows

- Built compliance data modules; income-limit import pipeline processing 1.6M HUD/Tax Credit records across 5 years; AWS S3 migration; 56 state-specific document templates; database schema redesign migrating hundreds of thousands of records with post-deployment stabilization.

PROJECTS

AgentProbe: Automated Data Quality Framework -- Reporting, Validation & Regression Detection

Data Quality · Automated Testing · Reporting · REST API · React · CI/CD

Python · FastAPI (REST API) · React.js · TypeScript · GitHub Actions · PostgreSQL | github.com/venkatalakshmiparimala/AgentProbe

Automated data quality and reporting framework: 5-metric validation engine tracking correctness, accuracy, and consistency across large datasets; CI-integrated regression detection alerting on data quality degradation; structured reporting dashboards for stakeholder analysis; REST API backend with React/TypeScript frontend. Demonstrates the data governance, validation, and reporting discipline this role requires.

CrashClarity: Large-Scale Data Analysis & Reporting (104K Records)

Python · R · Tableau · Data Analysis · Reporting · Dashboard

Python · R · scikit-learn · Random Forest · Tableau · GitHub Pages | Portfolio

End-to-end data analysis of 104K+ records -- data cleaning, feature engineering, predictive modeling (Random Forest + Logistic Regression, 85%+ accuracy), and interactive Tableau dashboard deployed for public access. Demonstrates full analytics pipeline from raw data to stakeholder-ready reporting and visualization.

AI on FHIR: Regulated Data Platform -- HIPAA Compliance & Access Controls

Flask · REST API · HIPAA · RBAC · Data Security · Full Stack

Python · Flask · React · Docker · FHIR API | github.com/venkatalakshmiparimala/AI_on_FHIR

HIPAA-compliant data platform with RBAC access controls, REST API integration with external health data systems, and secure data retrieval across structured regulated records -- demonstrates data security, access management, and compliance-aware system design.

EDUCATION

M.S. Data Analytics Engineering -- George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering -- JNTU Anantapur — 2013--2017